



RESEARCH ARTICLE

Efficacy of different nutrient sources on growth, seed yield and quality of finger millet (*Eleusine coracana* L.) varieties

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Abstract

Finger millet (*Eleusine coracana* L.) is a climate-resistant crop, which is high in calcium and dietary fibre. However, its significance in terms of food security is declining, as decreasing yields discourage farmers from cultivating it. The field experiment was conducted at Shri Guru Ram Rai (SGRR) University, Dehradun during 2023–24 and 2024–25, using eight genotypes and five nutrient management treatments. The study was conducted in randomised block design with 8 varieties and 5 treatments to evaluate seed quality and growth performance and yield. The treatments included farmyard manure, humic acid, biofertiliser, nano-fertilisers and locally prepared liquid formulations known as Jeevamrut and Neemastra. The findings revealed significant differences among treatments and genotypes. Treatment T₄ with local genotypes V₆ and V₇ resulted in maximum germination and seedling vigour whereas humic acid (T₄) and farmyard manure (T₁) produced superior yield attributes. The interaction between nutrient sources and varieties resulted in significant synergistic impacts in measuring integrated and eco-friendly nutrient management practices. Furthermore, this research indicates that with proper management, it is possible to increase yield, quality of the seed and seed longevity thereby promoting the use of local seeds and supporting the sustainability of finger millet production.

Keywords: assays; climate resistant; humic acid; Jeevamrut; Neemastra

Introduction

One of the staple cereal crops cultivated in India and East Africa but only rarely in other regions is finger millet (*Eleusine coracana* L.) which is commonly referred to as a neglected yet hardy drought-resistant cereal crop. Consequently, it plays an important role in ensuring food and nutritional security (1). It is morphologically an erect annual grass belonging to the family Poaceae and is characterised by finger-like racemes bearing small, hard seeds. Physiologically, finger millet is a C₄ plant and can therefore fix carbon more efficiently than C₃ plants; consequently, its water-use efficiency is higher. As a result, it is well adapted to regions experiencing water scarcity (1). Finger millet grains contain approximately 350 mg of calcium per 100 g of grains along with protein, dietary fibre and polyphenols. Therefore, they are considered nutritionally vital for preventing malnutrition and promoting health compared with most other cereals (2). However, the area under cultivation and the production of finger millet are declining drastically, due to the takeovers of the maize and rice-based cropping systems in the fields. This trend threatens nutritionally secure and sustainable agriculture because finger millet contains considerably higher levels of calcium and dietary fibre than most other cereals (3, 4). Consequently, the importance of these millets in sustainable agriculture and food security has been widely recognised, as the Food and Agriculture

Organization (FAO) officially declared 2023 the International Year of Millets (5). Although millets are nutritious and environmentally friendly crops, they have long been categorised as neglected or underutilised because the Green Revolution, primarily promoted wheat and rice through agricultural policies (6, 7). India is the leading producer of finger millet and therefore makes a significant contribution to food and nutrition security across many states. Finger millet or ragi production in India was approximately 1.386 million tonnes (Mt) during 2023–24. Karnataka contributes more than 60 % of the country's total production, followed by Tamil Nadu, Uttarakhand, Maharashtra, Odisha, Andhra Pradesh and Jharkhand (2). Finger millet is traditionally cultivated under marginal growing conditions and its grains can be stored for several years while remaining viable with minimal storage losses (8).

The improvement of finger millet production and grain quality depends largely on nutrient management. Farmyard manure (FYM), vermicompost and poultry manure are among the numerous organic sources that improve soil fertility and maintain long-term soil health (3). The biofertilisers such as *Azotobacter* enhance nutrient uptake and physiological efficiency, while humic substances improve root function, plant growth and nutrient availability (9). The recent introduction of nanotechnology-based fertilisers i.e., nano-urea and nano-fertilisers seems to promise a way

out of this situation as foliar application of nano-urea alone has been reported by different researchers to produce yields equivalent to those obtained with the full recommended nitrogen application (10, 11). In addition to this, native liquid formulations such as Jeevamrut and bioformulations such as Neemastra have shown potential to enhance crop growth, which can be environmentally friendly alternatives to chemical inputs (12). Most previous studies have primarily focused on yield and nutrient uptake and with limited attention given to seed quality characteristics of finger millet varieties under long-term storage conditions. Furthermore, only a few studies have compared different nutrient sources including FYM, humic acid, biofertilisers, nano-fertilisers and indigenous organic formulations such as Jeevamrut and Neemastra within a single experiment. These research gaps are addressed in the present study by examining the effects of various nutrient sources on seed yield, seed quality and the storability of varieties of finger millet seed. The treatments included 100 % of the recommended dose of nitrogen (RDN) through FYM, 50 % RDN combined with foliar application of humic acid, biofertilisers, nano-urea and nano-fertilisers and the indigenous formulations Jeevamrut and Neemastra. The leading credible eco-friendly substitutes, aiming to reduce the use of chemical fertilisers without any compromise to the production and seed quality, characterise the horizon of this mission, which hence transcends yield and storability. Therefore, this research is seen as assisting in enhancing the local seed systems on issues related to high price of inputs and soil erosion.

Materials and Methods

Experimental site and seed material

Seeds of eight finger millet (*Eleusine coracana* L.) genotypes were procured from the Almora, Nainital, Kanatal, Gopeshwar and Barkot regions of Uttarakhand. The experiment was conducted at the Department of Seed Science and Technology, Shri Guru Ram Rai (SGRR) University, Pathribagh Campus, Dehradun, Uttarakhand, India, located at 30.31369° N latitude, 78.04054° E longitude and an altitude of 648 m. The soil of the experimental site was sandy loam in texture, with favourable physicochemical properties and a slightly acidic to neutral pH, suitable for crop growth. Meteorological data were recorded monthly for two years (2023–24 and 2024–25) for various parameters, including maximum and minimum temperatures, relative humidity (RH-I and RH-II), wind speed and rainfall. It was found that the year 2024–25 had slightly higher maximum temperatures, especially during the summer period, but similar minimum temperatures compared with those of 2023–24. Humidity levels were generally high during the monsoon season, particularly from July to September, suggesting a good moisture supply. However, rainfall patterns were quite different, with 2023–24 experiencing more rainfall, particularly during the monsoon months, while 2024–25 observed lower rainfall. Wind speeds were moderate in both years, with a slight increase during the pre-monsoon and summer seasons.

Experimental layout and treatment details

The experiment was laid out in a factorial randomised block design with 3 replications, comprising eight genotypes and five nutrient treatments, resulting in 40 treatment combinations and 120 plots. Each 1 m² plot consisted of two rows with a spacing of 25 × 15 cm, covering a gross area of 288 m². The treatments included 100 % RDN through FYM, 50 % RDN with foliar application of humic acid,

biofertiliser (*Azotobacter*), nano urea with nano fertiliser and Jeevamrut with Neemastra. Standard agronomic practices were followed uniformly across all plots.

Observations recorded for growth and yield traits

Growth and yield traits of finger millet were systematically recorded at different growth stages. Field emergence (%) was determined by calculating the proportion of seedlings that emerged to the total number of seeds sown in each plot. Plant height was measured at 30 days after sowing (DAS) and again at harvest, from the base of the plant to the tip of the tallest panicle, using five randomly selected plants per plot. Phenological observations included days to the 2-leaf stage, 50 % flowering, physiological maturity and final harvest, which were recorded as the number of days from sowing to the respective developmental stage. Yield attributes were assessed at crop maturity. The number of panicles per plant was counted from five tagged plants in each plot, while the number of seeds per panicle was determined by threshing and counting the seeds from five representative panicles. Test weight was recorded as the mass of 1000 clean seeds sampled from the bulk produce. Seed yield per plant (g) was calculated from the harvested and cleaned seeds of five sampled plants, whereas seed yield per plot (kg) was obtained by harvesting, threshing and weighing the produce from the net plot area.

Observations recorded for seed quality

This experiment was conducted to evaluate the impact of different nutrient sources on seed quality of finger millet germplasm in the laboratory of Department of Seed Science and Technology, SGRR University. The recorded observations were germination percentage, speed of germination, root length, shoot length, fresh weight (g), dry weight (g) and seedling biomass. Seed quality traits were assessed in the laboratory using standard germination test procedures. All the parameters were observed according to the standard protocols of ISTA (1929) and Abdul-Baki and Anderson (1973) (13, 14).

Statistical analysis

The data obtained from the investigation were analysed using R software and Microsoft Excel.

Results

Nutrient source mediated variations in growth and phenological traits of finger millet varieties

The pooled results revealed that nutrient sources significantly influenced the growth and phenological traits of finger millet. Field emergence was highest in T₅ (86.46) and lowest in T₁ (83.61), while T₀ recorded the tallest plants at both 30 DAS (15.45 cm) and harvest (77.59 cm). Early seedling establishment occurred in T₀ (8.17 days), whereas T₅ was delayed (8.73 days). Flowering ranged from 53.42 days (T₄) to 55.40 days (T₂), with maturity being shortest in T₁ (110.03 days) and longest in T₄ (113.27 days) (Table 1). Among the varieties, PRM 1 (V₁) had better field emergence (87.09), plant height (76.57 cm) and crop duration (115.89 days) whereas V₆ recorded the lowest values. The PRM 1 (8.06 days) was the quickest in early seedling establishment whereas V₆ was the slowest (8.89 days). Flowering ranged from 53.14 days (GPU 48) to 55.75 days (PRM 2) and maturity ranged from 109.45 days (V₅) to 114.11 days (PRM 1) (Table 2). There was a significant interaction between treatments and varieties for the growth and phenological characteristics of finger millet (Fig. 1). The T₀-V₁ exhibited the highest field emergence and plant growth,

Table 1. Effect of treatments on growth and phenological traits of finger millet (pooled mean over two years)

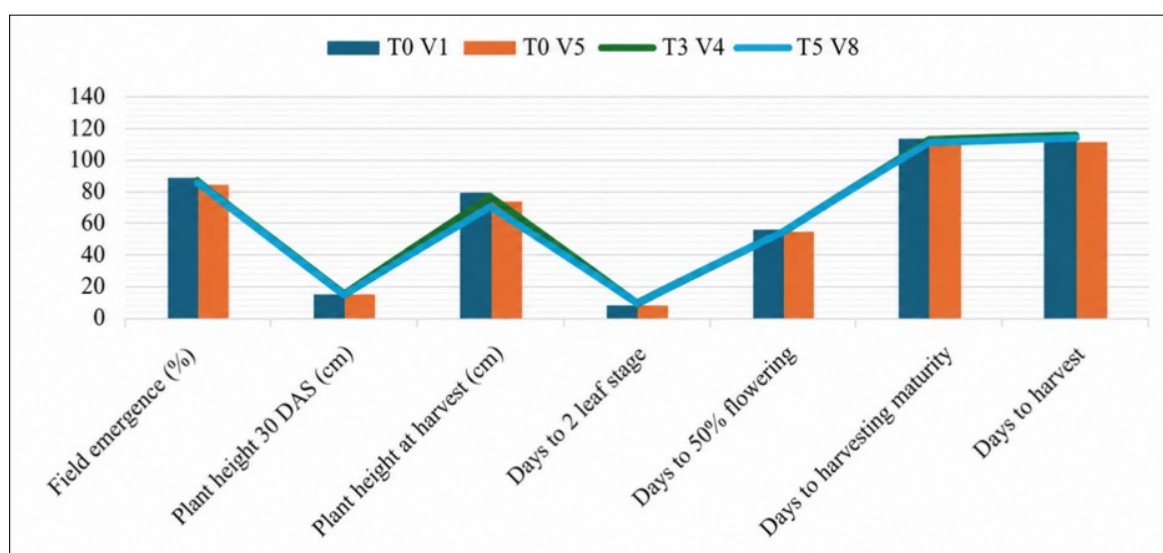
Treatment	Field Emergence	Plant height at 30 DAS (cm)	Plant height at harvest (cm)	Days to two leaf stage	Days to 50 % flowering	Days to harvesting maturity	Days to harvest
T ₀	85.56	15.45	77.59	8.17	53.92	110.13	111.81
T ₁	83.61	14.43	73.93	8.27	53.71	110.03	111.92
T ₂	84.67	13.97	72.02	8.67	55.40	113.08	115.06
T ₃	85.10	13.78	74.08	8.42	55.10	111.73	113.79
T ₄	84.62	13.71	73.49	8.60	53.42	113.27	115.36
T ₅	86.46	13.63	71.48	8.73	55.04	111.61	113.48
CD (5 %)	3.57	1.41	5.83	0.67	2.33	2.66	2.70
CV (%)	10.44	24.82	19.62	19.79	10.65	5.91	5.90

DAS: days after sowing.

Table 2. Effect of varieties on growth and phenological traits of finger millet (pooled mean over two years)

Treatment	Field emergence	Plant height at 30 DAS (cm)	Plant height at harvest (cm)	Days to two leaf stage	Days to 50 % flowering	Days to harvesting maturity	Days to harvest
V ₁ (PRM 1)	87.09	14.75	76.57	8.06	55.56	114.11	115.89
V ₂ (PRM 2)	83.75	14.47	74.99	8.25	55.75	110.14	112.03
V ₃ (VL 352)	85.70	14.41	75.04	8.61	54.06	111.87	113.89
V ₄ (GPU 48)	85.27	14.18	75.17	8.42	53.14	111.42	113.36
V ₅ (Local)	83.78	13.64	71.94	8.56	53.94	109.45	111.34
V ₆ (Local)	85.99	13.57	73.99	8.89	54.69	112.39	114.37
V ₇ (Local)	85.08	13.57	74.05	8.53	54.78	113.09	114.92
V ₈ (Local)	83.35	13.36	68.38	8.50	53.53	110.67	112.78
CD (5 %)	4.13	1.63	6.73	0.78	2.69	3.07	3.12
CV (%)	10.44	24.82	19.62	19.79	10.65	5.91	5.90

DAS: days after sowing.

**Fig. 1.** Interaction effect of treatments and varieties on growth and phenological traits of finger millet.

DAS: days after sowing.

indicating better crop establishment and growth. Differences in number of days to the 2-leaf stage, flowering, physiological maturity and harvesting among the treatment combinations indicated that the interactions between treatments and varieties had a particular effect on the crop development rate. Overall, the observed interaction indicates the significance of choosing the most effective treatment- variety combinations to enhance the performance of finger millet.

Impact of different nutrients sources on yield attributes of finger millet varieties

A pooled analysis of two-years showed that the nutrient sources as well as varietal differences had a significant impact on the yield performance of finger millet. The treatment T₁ produced the highest seed yield per plant (23.44 g) and T₄ produced the highest seed yield per plot (2.60 kg), both of which were superior to the control. The

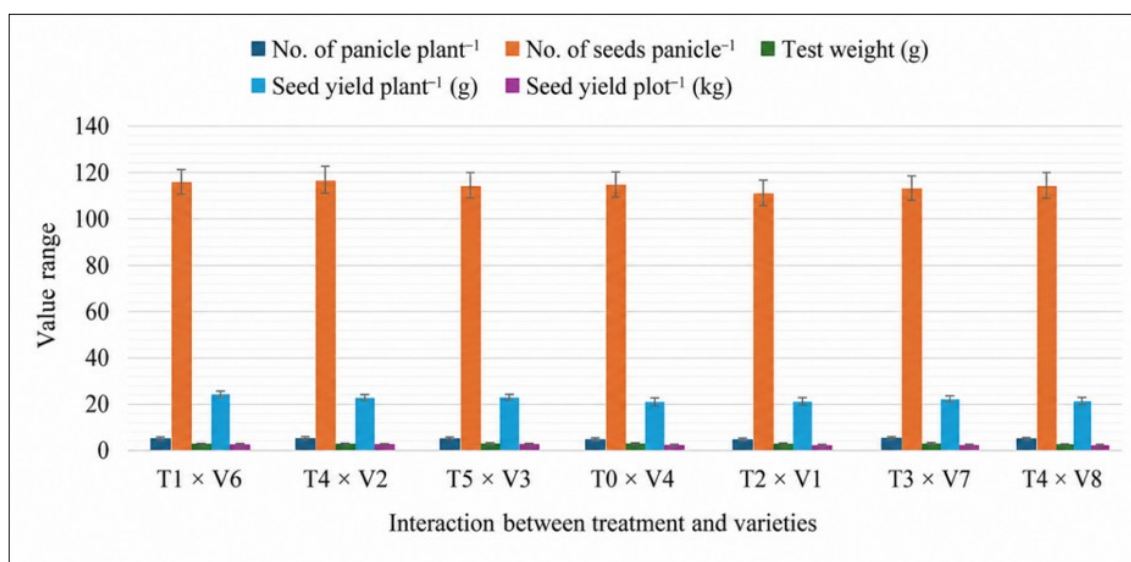
statistical results indicated that treatments T₃ and T₅ were statistically at par and T₂ had the lowest yield (Table 3). Varietal analysis revealed that V₆ (Local) had the highest seed yield per plant (23.15 g), followed by V₅ (22.90 g) and V₁ (22.81 g) and V₈ recorded the lowest yield (21.03 g). The V₂ (PRM 2) recorded the highest seed yield per plot (2.66 kg), whereas V₆ (2.59 kg) and V₃ (2.55 kg) did not differ significantly, but V₅ recorded the lowest yield (2.37 kg) (Table 4). The interaction effects also highlighted the complementary responses with T₁ × V₆ recording the highest seed yield per plant (~24–25 g) and T₄ × V₂ having the highest seed yield per plot (~2.70 kg) both of which were statistically at par with T₁ × V₆ and T₅ × V₃. The results underline that nutrient management significantly enhanced the productivity of finger millet under the nutrient management regimes and that varietal choice also influenced yield performance, with the T₁ and T₄ nutrient regimes and varieties V₆ and V₂ showing consistently good performance across the treatments (Fig. 2).

Table 3. The effect of different nutrient sources on yield attributes and seed yield of finger millet (pooled data of two years)

Treatment	No. of panicle plant ⁻¹	No. of seeds panicle ⁻¹	Test weight (g)	Seed yield plant ⁻¹ (g)	Seed yield plot ⁻¹ (kg)
T ₀	4.69	119.90	2.66	21.99	2.50
T ₁	5.34	110.94	2.84	23.44	2.56
T ₂	5.11	109.77	2.77	21.06	2.49
T ₃	5.11	115.44	2.85	22.52	2.41
T ₄	5.15	113.54	2.80	22.60	2.60
T ₅	5.17	113.15	2.71	21.89	2.57
CD (5 %)	0.61	7.87	0.17	1.72	0.22
C.V. (%)	29.67	17.16	15.40	19.15	21.74

Table 4. The effect of different varieties on yield attributes and seed yield of finger millet (pooled data of two years)

Treatment	No. of panicle plant ⁻¹	No. of seeds panicle ⁻¹	Test weigh (g)	Seed yield plant ⁻¹ (g)	Seed yield plot ⁻¹ (kg)
V ₁ (PRM 1)	4.89	112.11	2.79	22.81	2.52
V ₂ (PRM 2)	5.28	115.03	2.67	22.50	2.66
V ₃ (VL 352)	5.56	113.23	2.81	21.15	2.55
V ₄ (GPU 48)	4.75	116.23	2.82	22.14	2.44
V ₅ (Local)	4.92	110.17	2.75	22.90	2.37
V ₆ (Local)	5.31	115.50	2.81	23.15	2.59
V ₇ (Local)	4.89	113.42	2.83	22.31	2.53
V ₈ (Local)	5.14	114.64	2.70	21.03	2.52
CD (5 %)	0.70	9.08	0.20	1.98	0.25
C.V. (%)	29.67	17.16	15.40	19.15	21.74

**Fig. 2.** Effect of treatment × variety interaction on yield and yield attributes in finger millet.

Seed quality responses of finger millet varieties to different nutrient sources

There was high variation in the seed quality attributes of finger millet seeds across the treatments and varieties. The speed of germination was highest in T₄ (16.21), root length was greatest in T₄ (10.52 cm) and shoot length was highest in T₅ (15.71 cm), while the germination percentage ranged from 84.08 % (T₁) to 85.89 % (T₅). Fresh and dry weights varied slightly, with T₃ showing the highest fresh weight and T₄ the highest dry weight. The vigour index was notably higher in T₄ (771.52) than in the control, indicating its superiority (Table 5). Among the varieties, the germination percentage was highest in V₁ (PRM 1) and V₆ (Local) at 86.39 %, the speed of germination peaked in V₇ (15.63), root length was greatest in V₆ (10.32 cm) and shoot length was highest in V₈ (16.01 cm). Fresh weight was highest in V₁ (2.33 g), dry weight in V₅ (0.77 g) and the vigour index was highest in V₇ (774.94) (Table 6). According to Fig. 3, the interaction between treatment T₄ and local varieties V₆ and V₇ produced the highest values for most seed quality traits, including germination percentage, speed of germination, root and shoot length, fresh and dry weight, seedling biomass and vigour index. In particular, T₄ × V₆

(Local) showed slightly superior vigour index and related parameters, followed closely by T₄ × V₇ (Local), indicating that this treatment-variety combination was the most effective for enhancing seedling vigour and overall seed quality in finger millet. Other combinations under T₄ also performed well but were comparatively lower in vigour index and biomass, emphasising the synergistic benefit of using T₄ with local genotypes.

Discussion

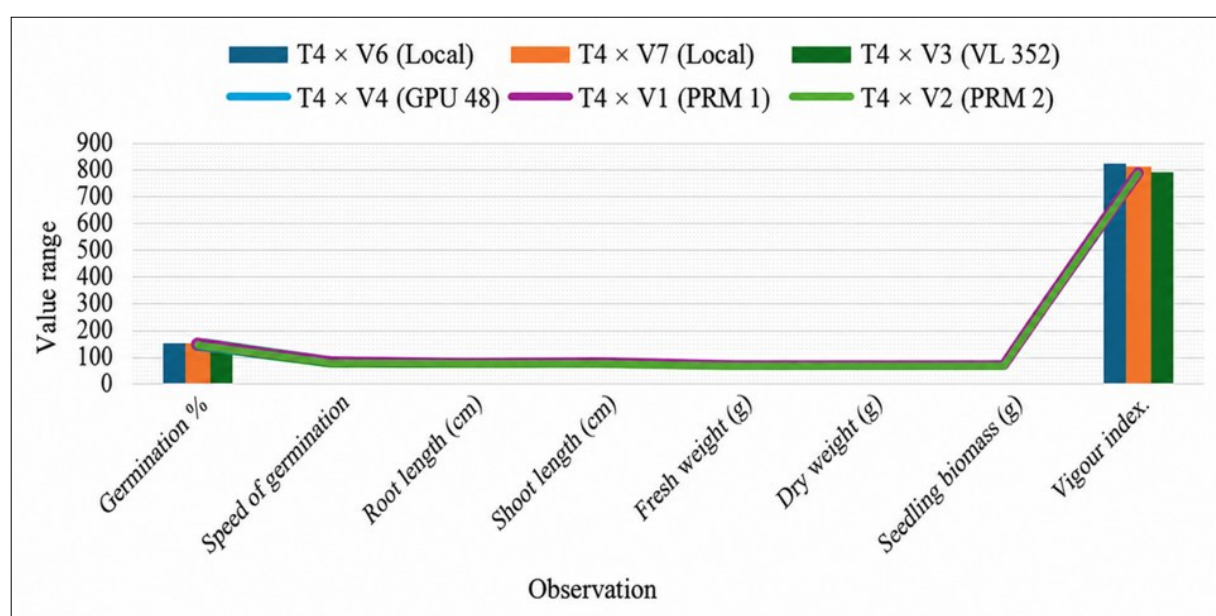
The combined two-year data provide a clear picture that the nutrient management practices adopted since 2021 have been a decisive factor influencing the growth and yield performance of finger millet and the integrated and organic nutrient management practices have played a significant role in generating noticeable changes in various characteristics. Fertilisers with organic components like FYM promoted early growth, plant height and dry matter accumulation, which is in line with the previous findings (15). Likewise, site-specific and integrated nutrient management also led to a significant enhancement of early vigour and overall productivity whereas

Table 5. The effect of different treatments on seed quality parameters of finger millet (pooled data of two years)

Treatment	Germination %	Speed of germination	Root length	Shoot length	Fresh weight (g)	Dry weight (g)	Seedling biomass	Vigor Index
T ₀	85.37	14.70	9.41	14.57	2.30	0.71	1.71	722.44
T ₁	84.08	14.82	9.30	15.19	2.30	0.72	1.71	743.80
T ₂	84.78	14.76	9.70	14.97	2.24	0.68	1.79	764.56
T ₃	85.07	14.97	9.82	14.86	2.36	0.71	1.69	763.81
T ₄	84.75	16.21	10.52	14.59	2.19	0.73	1.77	771.52
T ₅	85.89	15.40	9.78	15.71	2.16	0.67	1.80	758.63
CD (5 %)	2.39	1.18	1.17	1.14	0.19	0.069	0.16	60.00
C.V. (%)	6.97	19.36	29.68	18.95	20.65	24.28	23.36	19.76

Table 6. The effect of different genotypes on seed quality parameters of finger millet (pooled data of two years)

Treatment	Germination %	Speed of germination	Root length	Shoot length	Fresh weight (g)	Dry weight (g)	Seedling biomass	Vigour index
V ₁ (PRM 1)	86.39	15.06	10.04	15.10	2.33	0.66	1.66	727.94
V ₂ (PRM 2)	84.17	15.40	9.39	14.05	2.18	0.73	1.68	733.30
V ₃ (VL 352)	85.47	15.15	9.80	15.86	2.29	0.68	1.71	773.88
V ₄ (GPU 48)	85.18	15.18	10.00	14.90	2.27	0.73	1.87	764.64
V ₅ (Local)	84.19	14.92	9.81	14.90	2.24	0.77	1.71	761.12
V ₆ (Local)	86.39	15.06	10.32	14.60	2.30	0.70	1.90	774.43
V ₇ (Local)	85.67	15.63	8.68	14.44	2.24	0.69	1.64	774.94
V ₈ (Local)	85.06	14.76	9.99	16.01	2.21	0.67	1.80	722.76
CD (5 %)	2.76	1.36	1.35	1.32	0.22	0.079	0.19	69.29
C.V. (%)	6.97	19.36	29.68	18.95	20.65	24.28	23.36	19.76

**Fig. 3.** Interaction effect of treatment T₄ with finger millet varieties on seed quality parameters.

nutrient-stressed plots (equivalent to T₀) exhibited only initial elongation but failed to sustain growth throughout the crop cycle (16). Superior genotypes like PRM 1 and MR-6 exhibited better nutrient-use efficiency, showing enhanced responsiveness with high stover biomass, nitrogen uptake and grain productivity under integrated nutrient management regimes, which is consistent with the genotype $\times$ nutrient interaction studies (17). Similar improvements were observed in yield attributes, with integrated nutrient management producing significantly higher seed yield per plant and total plot yield than the unfertilised control.

In the present study, T₁ recorded the maximum seed yield per plant, while T₄ achieved the highest plot yield, both exceeding the control values, reflecting trends by similar reports (17–19). Varietal responses further emphasised that the local variety V₆ excelled in seed yield per plant, whereas the improved variety V₂ outperformed the others in plot-level yield, indicating genotype-specific nutrient management.

This is in line with earlier findings which, emphasised that the better varieties are more productive when the nutrient levels are under optimal regimes and some local ones were competitive under sustainable nutrient inputs (20, 21). The interaction effects that were observed, where T₁ $\times$ V₆ and T₄ $\times$ V₂ gave the highest yields, strengthen the genotype $\times$ nutrient management strategy synergy to maximise productivity (22). Beyond yield, integrated nutrient management is also associated with better soil health and nutrient availability which have a direct impact on grain weight, seeds per head and sustainable biomass production (17, 23). The twin advantages of improved growth and enhanced yield elements in the present study are very strong indicators of integrated nutrient solutions because they are the best method of growing finger millet. The pooled evidence supports that growth and yield of finger millet are greatly improved when the nutrient management regimes are combined with superior genotypes. Aligning physiological efficiency and nutrient inputs with varietal requirements enhances nutrient uptake and yield expression and makes a contribution towards

sustainable intensification. Such genotype $\times$ nutrient interactions not only optimise productivity, but also lead to resource-use efficiency, resilience and long-term sustainability of finger millet farming. The significant differences observed in seed quality characteristics between finger millet treatments and varieties fit prior research on the importance of both genetic and agronomic influences on seed performance. The increased rate of germination and root length observed in the present study is supported by the increases in root length and vigour index in treatment T₄. It is reported that early seedling vigour and optimised seed treatments could be used to improve seedling vigour by mitigating oxidative stress during seed storage and enhancing the metabolic processes involved in seedling activity during germination (24). Equally, local genotypes V₆ and V₇ were superior in root length, seedling biomass and vigour index. The genetic contribution to seed quality is verified by the characteristics identified as showing large genotypic differences in finger millet for critically important growth and vigour parameters (25). The greatest percentage of germination and fresh weight in V₁ also indicate that some superior lines retain greater initial seedling vigour, findings that suggest better cultivars exhibit superior performance under controlled conditions. The significant interaction effect indicates that some genotype-treatment combinations, particularly local genotypes with T₄ treatment, synergistically enhance seed quality traits, reinforcing the importance of integrated approaches combining genotype selection and effective seed treatment (26). This observation correlates with genome-wide association studies which have emphasised the polygenic nature of seed vigour and longevity characteristics that are determined by genetic background and environmental or treatment conditions (27). The optimisation of these interactions can significantly enhance seed storability, vigour and crop establishment under different environmental conditions.

Conclusion

The present investigation shows the important contribution of integrated and environmentally friendly nutrient management practices in improving the growth, productivity and seed quality of finger millet (*Eleusine coracana* L.). The findings clearly show that the nutrient treatments had a significant impact on crop performance, with T₁ (farmyard manure) being the most effective for seed yield per plant, while T₄ (humic acid-based treatment) was the best in terms of seed quality traits, such as germination, vigour index and biomass. Interestingly, the interaction effect between nutrient regimes and genotypes showed that local genotypes (V₆ and V₇) responded better to T₄ treatment, with the highest values for vigour index and other seed quality parameters. This suggests the significance of genotype $\times$ nutrient interactions in improving productivity and seed quality. In addition, the improved physiological efficiency and seed quality traits under integrated nutrient sources indicate their potential to contribute to improved stand establishment and seed longevity. Overall, the integrated use of organic fertilisers and novel nutrient sources, complemented by appropriate genotypes, provides an eco-efficient strategy to enhance the productivity and quality of finger millet seed. These approaches can provide robust agricultural systems, reduce reliance on chemical fertilisers and improved local seed systems, especially in resource-limited and climate-prone environments.

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Authors' contributions

AD performed data collection and drafted the manuscript. AT contributed to data collection. PK and GC participated in drafting the manuscript. VC performed the statistical analysis. S contributed to drafting and finalising the manuscript. All authors read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest: Authors do not have any conflict of interests to declare.

Ethical Issue: None

References

- Gupta SM, Arora S, Mirza N, Pande A, Lata C, Puranik S, et al. Finger millet: a "certain" crop for an "uncertain" future and a solution to food insecurity and hidden hunger under stressful environments. *Front Plant Sci.* 2017;8:643. <https://doi.org/10.3389/fpls.2017.00643>
- Agricultural and Processed Food Products Export Development Authority. Indian millets: finger millet (Ragi). New Delhi: APEDA; 2024.
- Thilakarathna MS, Raizada MN. A review of nutrient management studies involving finger millet in the semi-arid tropics of Asia and Africa. *Agronomy.* 2015;5(3):262–90. <https://doi.org/10.3390/agronomy5030262>
- Triveni U, Rani SY, Patro TSSK, Anuradha N, Divya M. Fertilizer responsiveness of short duration improved finger millet genotypes to different levels of NPK fertilizers. *Indian J Agric Res.* 2018;52(1):97–100. <https://doi.org/10.18805/IJARE.A-4801>
- Muskan F, Redhu M, Redhu S, Rahimi M. Millets in the global market: a critical review of challenges and opportunities. *Food Prod Process Nutr.* 2025;7:37. <https://doi.org/10.1186/s43014-025-00316-1>
- Mohan MM, Vijayakumar A, Bang-Berthelsen CH, Mudnakudu-Nagaraju KK, Shetty R. Millets: journey from an ancient crop to sustainable and healthy food. *Foods.* 2025;14(10):1733. <https://doi.org/10.3390/foods14101733>
- Purohit S, Palai B. Finger millet: a boon to food security under changing climatic condition. *Int J Chem Stud.* 2020;8(3):1385–8. <https://doi.org/10.22271/chemi.2020.v8.i3s.9390>
- MilletStats. Finger millet (Ragi) statistics in India. 2018.
- Yadav B, Bharati V, Yadav TK, Kumar A, Kumar A, Nath S, et al. Impact of liquid biofertilizers on nutrition of finger millet and residual soil fertility status. *J Exp Agric Int.* 2025;47(4):314–22. <https://doi.org/10.9734/jeai/2025/v47i43380>
- Samanta S, Maitra S, Shankar T, Gaikwad D, Sagar L, Panda M, et al. Comparative performance of foliar application of urea and nano urea on finger millet (*Eleusine coracana* L. Gaertn). *Crop Res.* 2022;57(3):166–70. <https://doi.org/10.31830/2454-1761.2022.025>
- Soundarya S, Gaddi AK, Veeresh H, Ravi S, Srinivasa DK, Ajaya Kumar MY. Effect of nano urea on growth, yield and nutrient uptake of finger millet (*Eleusine coracana* L.). *Int J Plant Soil Sci.* 2024;36(9):660–7. <https://doi.org/10.9734/ijpss/2024/v36i95014>

12. Srilekha GV. Effect of lime, compost and micronutrients on crop yield and seed quality of finger millet. MSc (thesis). Bhubaneswar: Odisha University of Agriculture and Technology; 2023.
13. International Seed Testing Association. International rules for seed testing. Zürich: International Seed Testing Association; 1929.
14. Abdul-Baki AA anderson JD. Vigor determination in soybean seed by multiple criteria. Crop Sci. 1973;13(6):630–3. <https://doi.org/10.2135/cropsci1973.0011183X001300060013x>
15. Aparna NS, Karthikeyan PG. Effect of integrated nutrient management on growth and yield of finger millet (*Eleusine coracana* (L.) Gaertn). Mysore J Agric Sci. 2023;57(3):266–75.
16. Patel VD, Mishra R, Soni R. Effect of nutrient management on growth and yield of finger millet in hilly area. Pharma Innov J. 2023;12(10):234–9.
17. Prabhakar M, Dass S, Pramanick B, et al. Potential for yield and soil fertility improvement with nutrient management practices in finger millet. Front Nutr. 2023;10:1095449. <https://doi.org/10.3389/fnut.2023.1095449>
18. Sivakumar T, Pushpa J, Kumar R. Nutrient management practices on growth and yield attributes of finger millet (*Eleusine coracana*). Int J Environ Clim Change. 2023;13(4):125–33.
19. Bhalerao GL, Sawargaonkar SL, Bagewadi BV. Bundled management practices for enhanced finger millet productivity and resource use efficiency in semi-arid tropics. Plant Sci Today. 2025;12(8):650–62.
20. Teklu D, Gashu D, Joy EJM, Bailey EH, Wilson L, Amede T, et al. Differences in the nutritional quality of improved finger millet genotypes in Ethiopia. Sci Rep. 2024;14(1):460. <https://doi.org/10.1038/s41598-023-48749-3>
21. Anitha S, Gowda H, Manjunatha MK, Kumar MS. Variation in the nutrient content of different genotypes and nutrient management practices in finger millet. Front Sustain Food Syst. 2024;8:1324046. <https://doi.org/10.3389/fsufs.2024.1324046>
22. Pramanick B, Dass S, et al. Can site-specific nutrient management improve the productivity of finger millet? Heliyon. 2024;10(6):e08805. <https://doi.org/10.1016/j.heliyon.2024.e32774>
23. Ebanyat P, Wendt J, Šimůnek J, et al. Efficacy of nutrient management options for finger millet production under rainfed conditions. Front Sustain Food Syst. 2021;5:674926. <https://doi.org/10.3389/fsufs.2021.674926>
24. Kannababu N, Srikanth S, Kumar R. Determining seed longevity and vigor in finger millet. Int J Genet Plant Breed. 2024;8(9):90–98.
25. Amisha N, Vijayakumari K, Tirumala K. Evaluation of germplasm lines for seed quality traits in finger millet. J Exp Agric Int. 2020;47(2):237–48. <https://doi.org/10.9734/jeai/2025/v47i23283>
26. Vijayakumari K, Maya S, Ramakrishnan N. Seed quality traits of finger millet varieties under controlled conditions. Indian J Agric Sci. 2003;73(12):672–6.
27. Kannababu N, Srikanth S, Kumar R. Role of functional genes for seed vigor related traits in finger millet. Sci Rep. 2025;15:89315. <https://doi.org/10.1038/s41598-025-02029-4>

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